



Rating
Hold

North America
United States

TMT
Software

Company
Check Point

Reuters
CHKP.OQ

Bloomberg
CHKP US

Date
30 July 2026

Rating	Hold
Price target (USD)	130.00
Price at 29 Jul 26	139.67
52-week range	206.91 – 112.47

[Valuation & Risks](#)

2Q - A business still in transition

Check Point reported a mixed quarter as modest billings and Product beats were overshadowed by Service-driven revenue underperformance. With growth largely reliant on hardware firewall refresh within its install base, the company is grappling with headwinds across revenue segments as the Quantum Force cycle reaches its conclusion and is reflected in maintenance and professional services weakness, y/y unit declines, and minimal evidence of pricing power at a time when peer FTNT is noting high-single-digit contribution from price increases. Positively, 2026 guidance was maintained across the board, though seasonally more weighted to 4Q than Street modeled as mgmt. saw some "large appliance deals that were pushed from Q3 to Q4". This naturally creates a tough setup for 4Q with guidance embedding ~\$800m of Revenue in the quarter (the first time ever) and a return to Product growth which implies ~flat y/y unit growth, assuming minimal pricing contribution and no mix change (shift to smaller appliances has pressured Product GM this year) - no mean feat at the tail end of a hardware refresh cycle.

That all said, mgmt. remains focused on catalyzing long-term growth reacceleration. After a GTM reorganization in 1Q, including appointing a new CRO, CHKP is planning to hire ~300 quota-carrying reps (100-150 net) in 2H. As we have noted following similar announcements from cyber peers, ramping reps is not without risk, not least within competitive, increasingly saturated markets (i.e., SASE). We are inclined to give new CRO Sherif Seddik time to implement his strategy, and we don't expect meaningful new business contribution from these new reps until 2027 at the earliest. That said, with underlying NGOM (i.e., exc. R&D tax credit) now <35% (down >600bps y/y), we also ponder the impact that adding 15-20% more quota-bearing reps may have on CHKP's immediate and long-term profitability.

Emerging products continue to be major growth drivers as email security, ERM, and AI security in aggregate grew ARR >40% and billings >35%. Sustained momentum with the group underwrites mgmt.'s confidence in reiterating Security Subscriptions Revenue guidance for the year despite attached subscription weakness. Ultimately, though, this remains a business in transition as mgmt. works to position CHKP as a security partner of choice for the AI era. We therefore remain on the sidelines until we see clearer green shoots of sustainable top-line reacceleration. Maintain Hold rating and DCF-derived TP of \$130.

What we liked / what we're watching on the quarter

(=) 2Q total calculated billings of \$639m declined y/y (-0.5%) for a consecutive quarter though slimly surpassed Street (\$637.5m/-0.8% y/y). That said, upside

Deutsche Bank Securities Inc.

IMPORTANT RESEARCH DISCLOSURES AND ANALYST CERTIFICATIONS LOCATED IN APPENDIX 1. Deutsche Bank does and seeks to do business with companies covered in its research reports. Thus, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.



was entirely driven by duration (non-current deferred revenue +\$10m/+1.6% vs Street) following Revenue underperformance vs the midpoint of guidance and lower current deferred revenue vs consensus estimates (-\$5m/-0.3%). This is the fifth consecutive quarter current deferred revenue has underperformed Street. Put differently, current calculated billings and current calculated service billings missed implied consensus by -\$6m/-1.0% and -\$10m/-1.9%, respectively.

(-) 2Q Revenue of \$673.6m/+1.3% y/y missed the guidance midpoint (\$675m/+1.5% y/y) for a consecutive quarter. Underperformance was driven by another quarter of Service Revenue underperformance, including a modest (-\$0.4m) miss vs the midpoint of Security Subscriptions guidance and a more notable -\$7m/-2.9% miss vs Street Software Updates, Maintenance & Services.

(-) Product Revenue declined -14% y/y from -3% y/y in 1Q as the Quantum Force cycle reaches its conclusion. Pricing was a minimal benefit in the quarter as most of the list increase was cannibalized by elevated discounting.

(-) 2Q NGOM of 38.6% fell -220bps y/y on a headline basis and came in lower than consensus (39.0%) for the seventh consecutive quarter. Together with the top-line underperformance, NGOI missed Street by -\$3m/-1.3%. The benefit from the Israeli R&D tax credit was a touch greater in 2Q than in 1Q on a \$ basis (\$28m vs \$27m), and underlying margins continue to deteriorate - excluding this credit, NGOM came in at 34.4%, -640bps y/y.

(=) RPO grew 7% y/y to \$2,552m, accelerating ~10bps sequentially albeit against a ~540bp easier compare. Together with Total Revenue, calculated total RPO bookings returned to growth (+1.0% y/y from -2.4% y/y) after declining in 1Q, albeit against a ~14pt easier compare.

(+) Emerging products growth remains solid as combined Email Security / ERM / AI Security ARR growth exceeded 40% y/y, while combined billings grew >35% y/y. That said, we note that SASE was removed in 2Q and replaced with AI Security as growth for the product fell below the group's aggregate growth.

(+) \$325m of shares were repurchased in the quarter, in line with the usual run rate.

[What we liked / what we're watching on the guide](#)

(=) 2026 Revenue guidance was maintained, though 2H Revenue guidance is more weighted to 4Q than Street was modeling to reflect some large appliance deals pushing from 3Q. Product growth is expected to be similar to 2Q (-14% y/y) and turn positive in 4Q. Subscription Revenue growth is likewise guided to decelerate in 3Q before notably reaccelerating in 4Q. This leaves a tricky 4Q setup, implying ~\$800m Revenue in the quarter - the highest on record.

(=) Despite incremental hiring plans in 2026, the full year adjusted free cash flow and EPS guides were maintained.

[Estimate Changes and Valuation](#)

Incorporating 2Q results, guidance, and our updated thoughts on the business, we adjust our 2026/2027 revenue estimates to \$2,810m/\$2,992m from \$2,811m/\$3,020m and our 2027 non-GAAP EPS to \$11.50 from \$11.79. We maintain our Hold rating and our DCF-derived TP of \$130. Our DCF is based on a WACC of 9.3%, a terminal risk-free rate of 4%, equity risk premium of 5.25% and



terminal growth rate of 3%. Upside risks include: (1) Acquisitions and improved execution under CEO Nadav Zafrir drive top-line and Billings outperformance, (2) Greater-than-expected new logo acquisition within core firewall, & (3) Beneficiary of vendor consolidation enabling greater cross-sell of Harmony and cloud network security. Downside risks include: (1) Continued difficult backdrop (i.e., rising component prices) resulting in customers sweating existing appliances to preserve cash, (2) failure to see productive returns on S&M investments, and (3) Increased FW displacements from PANW and FTNT.



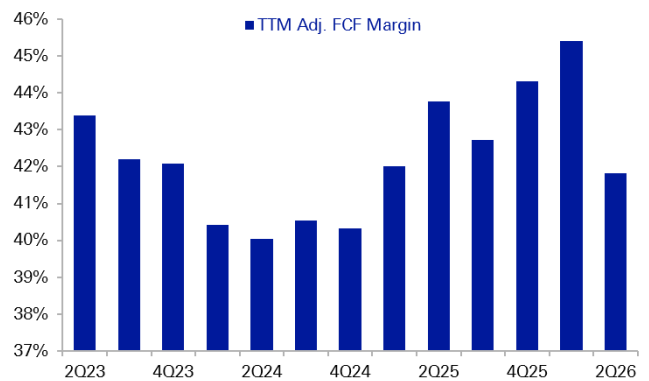
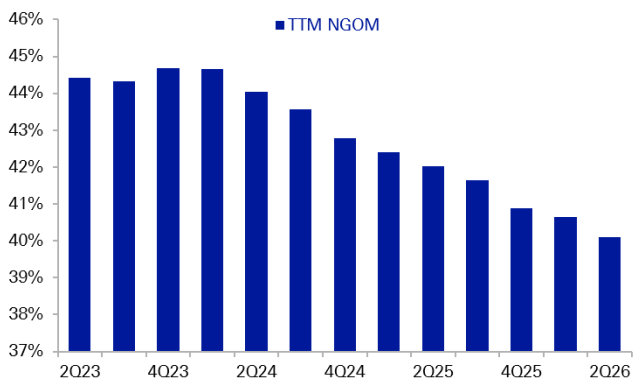
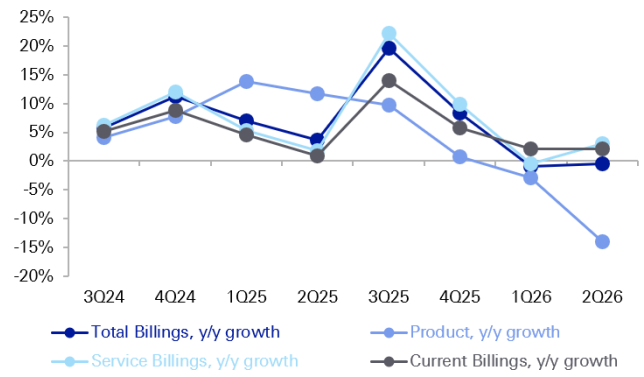
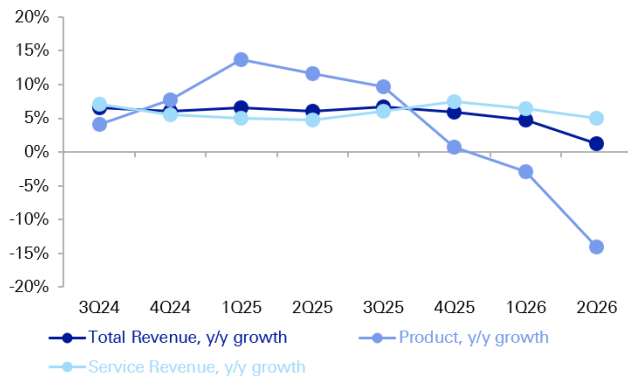
Figure 1: 2Q results

Check Point (CHKP)						
\$mn (Non-GAAP unless otherwise stated)						
	2Q26					
	Actual	DBe	vs DBe	Street	vs Street	Guidance
Income Statement						
Product	113	106	6.9%	109	3.7%	
Y/Y Growth	-14.0%	-19.6%		-17.1%		
Q/Q Growth	2.3%	-4.3%		-1.4%		
Security Subscriptions	333	333	-0.1%	333	-0.1%	333
Y/Y Growth	11.6%	11.8%		11.8%		
Q/Q Growth	2.9%	3.0%		3.0%		
Software Updates, Maintenance and Services	228	236	-3.6%	234	-2.9%	
Y/Y Growth	-3.3%	0.3%		-0.4%		
Q/Q Growth	-2.9%	0.7%		0.0%		
Total Revenue	674	675	-0.2%	675	-0.2%	675
Y/Y Growth	1.3%	1.5%		1.5%		
Q/Q Growth	0.8%	1.0%		1.0%		
Gross Profit	588	594	-1.0%	594	-1.0%	
Gross Margin	87.3%	88.0%	-70bps	88.0%	-71bps	
Operating Profit	260	264	-1.7%	263	-1.3%	
Operating Margin	38.6%	39.1%	-58bps	39.0%	-45bps	
Net Income	264	255	3.2%	257	2.5%	
Diluted EPS	2.55	2.45	3.9%	2.45	3.9%	2.45
Operating Metrics						
Service Billings	526	530	-0.8%	527	-0.2%	
Y/Y Growth	3.0%	3.8%		3.2%		
Q/Q Growth	20.4%	21.4%		20.6%		
Total Billings	639	636	0.5%	636	0.5%	
Y/Y Growth	-0.5%	-1.0%		-1.0%		
Q/Q Growth	16.8%	16.2%		16.2%		
Operating Cash Flow	170	167	1.7%	167	1.4%	
Margin	25.2%	24.7%		65.1%		
Adj. Free Cash Flow	161	160	0.5%	160	0.4%	160
Margin	23.9%	23.7%		62.3%		

Source: Company Reports, Bloomberg Finance L.P., Deutsche Bank Research



Figure 2: Key charts



Source: Company Reports, Deutsche Bank Research



Figure 3: Estimate Changes

Check Point (CHKP)

\$mn (Non-GAAP unless otherwise stated)

	3Q26				FY26			
	New Est.	Old Est.	Change	Guidance	New Est.	Old Est.	Change	Guidance
Income Statement								
Product	112	116	-3.4%		510	494	3.3%	
Y/Y Growth	-14.0%	-11.0%			-6.9%	-9.9%		
Q/Q Growth	-1.1%	9.5%			0.0%	0.0%		
Security Subscriptions	338	343	-1.6%	338	1,365	1,365	0.0%	1,365
Y/Y Growth	10.5%	12.3%			12.0%	12.0%		
Q/Q Growth	1.5%	3.0%			0.0%	0.0%		
Software Updates, Maintenance and Services	220	239	-7.9%		935	952	-1.8%	
Y/Y Growth	-8.8%	-1.0%			-2.5%	-0.7%		
Q/Q Growth	-3.2%	1.4%			0.0%	0.0%		
Total Revenue	670	698	-4.0%	670	2,810	2,811	0.0%	2,810
Y/Y Growth	-1.1%	3.1%			3.1%	3.1%		
Q/Q Growth	-0.5%	3.4%						
Gross Profit	586	615	-4.6%		2,452	2,467	-0.6%	
Gross Margin	87.5%	88.0%	-53bps		87.3%	87.8%	-52bps	
Operating Profit	256	281	-8.9%		1,088	1,115	-2.5%	
Operating Margin	38.3%	40.3%	-203bps		38.7%	39.7%	-96bps	
Net Income	253	269	-6.1%		1,076	1,078	-0.1%	
Diluted EPS	2.48	2.63	-5.8%	2.48	10.45	10.45	0.0%	10.45
Operating Metrics								
Service Billings	557	588	-5.3%		2,468	2,509	-1.6%	
Y/Y Growth	3.0%	8.7%			4.7%	6.4%		
Q/Q Growth	6.0%	11.0%			0.0%	0.0%		
Total Billings	670	705	-5.0%		2,978	3,003	-0.8%	
Y/Y Growth	-0.3%	4.9%			2.5%	3.3%		
Q/Q Growth	4.7%	10.7%			0.0%	0.0%		
Operating Cash Flow	257	222	15.8%		1,212	1,209	0.2%	
Margin	38.3%	31.8%			43.1%	43.0%		
Adj. Free Cash Flow	250	215	16.5%	250	1,200	1,200	0.0%	1,200
Margin	37.3%	30.8%			42.7%	42.7%		

Source: Company Reports, Bloomberg Finance L.P., Deutsche Bank Research



Appendix 1

Important Disclosures

Company	Ticker	Recent price	Disclosure
Check Point	CHKP.OQ	139.67 (USD) 29 Jul 26	2, 7, 8, 14, 21, 24, 26

Prices are current as of the end of the previous trading session unless otherwise indicated and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Other information is sourced from Deutsche Bank, subject companies, and other sources.

Important Disclosures Required by U.S. Regulators

2 - Deutsche Bank and/or its affiliate(s) may act as a market maker or liquidity provider in the financial instruments issued by this company.

7 - Deutsche Bank and/or its affiliate(s) has received compensation from this company for the provision of investment banking or financial advisory services within the past year.

8 - Deutsche Bank and/or its affiliate(s) expects to receive, or intends to seek, compensation for investment banking services from this company in the next three months.

14 - Deutsche Bank and/or its affiliate(s) has received compensation from this company within the past year for non-investment banking related services.

21 - This company has been a client of Deutsche Bank Securities Inc. within the past year, during which time it received non-investment banking securities-related services.

Important Disclosures Required by Non-U.S. Regulators

2 - Deutsche Bank and/or its affiliate(s) may act as a market maker or liquidity provider in the financial instruments issued by this company.

24 - Deutsche Bank and/or its affiliate(s) is or has been over the previous 12 months party to an agreement with the company relating to the provision of services set out in Sections A and B of Annex I of Directive 2014/65/EU, or has over the previous 12 months been obliged or entitled (as applicable) to pay or receive compensation relating to the provision of services set out in Sections A and B of Annex I of Directive 2014/65/EU.

26 - Within the preceding 12 months, Deutsche Bank and/or its affiliate(s) has received compensation for the provision of investment banking services or is currently providing or has provided investment banking services to this company.

For disclosures pertaining to recommendations or estimates made on securities other than the primary subject of this research, please see the most recently published company report or visit our global disclosure look-up page on our website at <https://research.db.com/Research/Disclosures/EquityResearchDisclosures>. Aside from within this report, important risk and conflict disclosures can also be found at <https://research.db.com/Research/Disclosures/Disclaimer>. Investors are strongly encouraged to review this information before investing.

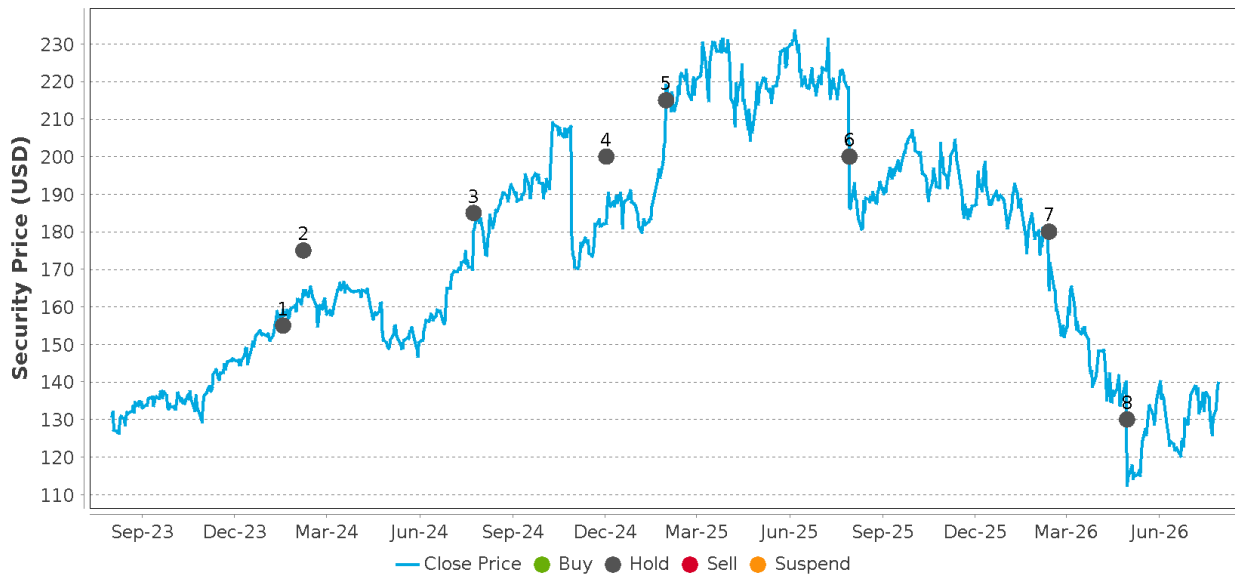
Analyst Certification

The views expressed in this report accurately reflect the personal views of the undersigned lead analyst(s). In addition, the undersigned lead analyst(s) has not and will not receive any compensation for providing a specific recommendation or view in this report. Brad Zelnick.





Historical recommendations and target price: Check Point (CHKP.OQ)



1.	18 Jan 24	Price target: 155 (USD), close price 159.06 (USD), Recommendation: Hold, Analyst: Brad Zelnick
2.	7 Feb 24	Price target: 175 (USD), close price 164.33 (USD), Recommendation: Hold, Analyst: Brad Zelnick
3.	24 Jul 24	Price target: 185 (USD), close price 180.07 (USD), Recommendation: Hold, Analyst: Brad Zelnick
4.	2 Dec 24	Price target: 200 (USD), close price 182.18 (USD), Recommendation: Hold, Analyst: Brad Zelnick
5.	30 Jan 25	Price target: 215 (USD), close price 218.99 (USD), Recommendation: Hold, Analyst: Brad Zelnick
6.	30 Jul 25	Price target: 200 (USD), close price 186.67 (USD), Recommendation: Hold, Analyst: Brad Zelnick
7.	12 Feb 26	Price target: 180 (USD), close price 164.47 (USD), Recommendation: Hold, Analyst: Brad Zelnick
8.	30 Apr 26	Price target: 130 (USD), close price 112.47 (USD), Recommendation: Hold, Analyst: Brad Zelnick



Company rating dispersion and banking relationships

DBSI Companies under Coverage	Buy	Hold	Sell
Covered	58%	42%	0%
w/ Banking relationship	42%	35%	0%
w/ MiFID services	63%	52%	67%

Global Companies under Coverage	Buy	Hold	Sell
Covered	58%	40%	2%
w/ Banking relationship	46%	32%	32%
w/ MiFID services	75%	69%	94%

Company Rating and Dispersion Key

The above table provides a snapshot of Deutsche Bank's company research rating distribution across our covered companies. We also present the percentage of companies where Deutsche Bank has provided Investment Banking Services in the past 12 months and/or MIFID Investment and Ancillary services, in the past 12 months. Please see the key and definition of our rating below.

Note - percentages are rounded so may not total 100%.

Covered: The overall rating distribution across all companies under coverage with a rating.

w/Banking relation: Percentage of companies under coverage with a rating within each of the "buy", "hold" and "sell" categories for which Deutsche Bank has provided Investment Banking Services within the previous 12 months.

w/MiFID services: Percentage of companies under coverage with a rating within each of the "buy", "hold" and "sell" categories for which Deutsche Bank has provided MIFID Investment and Ancillary services within the previous 12 months.

Buy/Hold/Sell Percentages: These percentages reflect the proportion of companies within each category that have been assigned the corresponding rating, based on our 12-month view of Total Shareholder Return (TSR).

Rating definitions:

Buy: Based on a current 12-month view of TSR, we recommend that investors buy the stock.

Sell: Based on a current 12-month view of TSR, we recommend that investors sell the stock.

Hold: We take a neutral view on the stock 12-months out and, based on this time horizon, do not recommend either a Buy or Sell.

TSR: Total Shareholder Return. Percentage change in share price from current price to projected target price plus projected dividend yield.

Newly issued research recommendations and target prices supersede previously published research.



Additional Information

The information and opinions in this report were prepared by Deutsche Bank AG or one of its affiliates (collectively 'Deutsche Bank'). Though the information herein is believed to be reliable and has been obtained from public sources believed to be reliable, Deutsche Bank makes no representation as to its accuracy or completeness. Hyperlinks to third-party websites in this report are provided for reader convenience only. Deutsche Bank neither endorses the content nor is responsible for the accuracy or security controls of those websites.

If you use the services of Deutsche Bank in connection with a purchase or sale of a security that is discussed in this report, or is included or discussed in another communication (oral or written) from a Deutsche Bank analyst, Deutsche Bank may act as principal for its own account or as agent for another person.

Deutsche Bank may consider this report in deciding to trade as principal. It may also engage in transactions, for its own account or with customers, in a manner inconsistent with the views taken in this research report. Others within Deutsche Bank, including strategists, sales staff and other analysts, may take views that are inconsistent with those taken in this research report. Deutsche Bank issues a variety of research products, including fundamental analysis, equity-linked analysis, quantitative analysis and trade ideas. Recommendations contained in one type of communication may differ from recommendations contained in others, whether as a result of differing time horizons, methodologies, perspectives or otherwise. Deutsche Bank and/or its affiliates may also be holding debt or equity securities of the issuers it writes on. Analysts are paid in part based on the profitability of Deutsche Bank AG and its affiliates, which includes investment banking, trading and principal trading revenues.

Opinions, estimates and projections constitute the current judgment of the author as of the date of this report. They do not necessarily reflect the opinions of Deutsche Bank and are subject to change without notice. Deutsche Bank provides liquidity for buyers and sellers of securities issued by the companies it covers. Deutsche Bank research analysts sometimes have shorter-term trade ideas that may be inconsistent with Deutsche Bank's existing longer-term ratings. Some trade ideas for equities are listed as Catalyst Calls on the Research Website (<https://research.db.com/Research/>), and can be found on the general coverage list and also on the covered company's page. A Catalyst Call represents a high-conviction belief by an analyst that a stock will outperform or underperform the market and/or a specified sector over a time frame of no less than two weeks and no more than three months. In addition to Catalyst Calls, analysts may occasionally discuss with our clients, and with Deutsche Bank salespersons and traders, trading strategies or ideas that reference catalysts or events that may have a near-term or medium-term impact on the market price of the securities discussed in this report, which impact may be directionally counter to the analysts' current 12-month view of total return or investment return as described herein. Deutsche Bank has no obligation to update, modify or amend this report or to otherwise notify a recipient thereof if an opinion, forecast or estimate changes or becomes inaccurate. Coverage and the frequency of changes in market conditions and in both general and company-specific economic prospects make it difficult to update research at defined intervals. Updates are at the sole discretion of the coverage analyst or of the Research Department Management, and the majority of reports are published at irregular intervals. This report is provided for informational purposes only and does not take into account the particular investment objectives, financial situations, or needs of individual clients. It is not an offer or a solicitation of an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Target prices are inherently imprecise and a product of the analyst's judgment. The financial instruments discussed in this report may not be suitable for all investors, and investors must make their own informed investment decisions. Prices and availability of financial instruments are subject to change without notice, and investment transactions can lead to losses as a result of price fluctuations and other factors. If a financial instrument is denominated in a currency other than an investor's currency, a change in exchange rates may adversely affect the investment. Past performance is not necessarily indicative of future results. Performance calculations exclude transaction costs, unless otherwise indicated. Unless otherwise indicated, prices are current as of the end of the previous trading session and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Data is also sourced from Deutsche Bank, subject companies, and other parties. Artificial intelligence tools may be used in the preparation of this material, including but not limited to assist in fact-finding, data analysis, pattern recognition, content drafting and editorial corrections pertaining to research material.



The Deutsche Bank Research Department is independent of other business divisions of the Bank. Details regarding our organizational arrangements and information barriers we have to prevent and avoid conflicts of interest with respect to our research are available on our website (<https://research.db.com/Research/>) under Disclaimer.

Macroeconomic fluctuations often account for most of the risks associated with exposures to instruments that promise to pay fixed or variable interest rates. For an investor who is long fixed-rate instruments (thus receiving these cash flows), increases in interest rates naturally lift the discount factors applied to the expected cash flows and thus cause a loss. The longer the maturity of a certain cash flow and the higher the move in the discount factor, the higher will be the loss. Upside surprises in inflation, fiscal funding needs, and FX depreciation rates are among the most common adverse macroeconomic shocks to receivers. But counterparty exposure, issuer creditworthiness, client segmentation, regulation (including changes in assets holding limits for different types of investors), changes in tax policies, currency convertibility (which may constrain currency conversion, repatriation of profits and/or liquidation of positions), and settlement issues related to local clearing houses are also important risk factors. The sensitivity of fixed-income instruments to macroeconomic shocks may be mitigated by indexing the contracted cash flows to inflation, to FX depreciation, or to specified interest rates - these are common in emerging markets. The index fixings may - by construction - lag or mis-measure the actual move in the underlying variables they are intended to track. The choice of the proper fixing (or metric) is particularly important in swaps markets, where floating coupon rates (i.e., coupons indexed to a typically short-dated interest rate reference index) are exchanged for fixed coupons. Funding in a currency that differs from the currency in which coupons are denominated carries FX risk. Options on swaps (swaptions) the risks typical to options in addition to the risks related to rates movements.

Derivative transactions involve numerous risks including market, counterparty default and illiquidity risk. The appropriateness of these products for use by investors depends on the investors' own circumstances, including their tax position, their regulatory environment and the nature of their other assets and liabilities; as such, investors should take expert legal and financial advice before entering into any transaction similar to or inspired by the contents of this publication. The risk of loss in futures trading and options, foreign or domestic, can be substantial. As a result of the high degree of leverage obtainable in futures and options trading, losses may be incurred that are greater than the amount of funds initially deposited - up to theoretically unlimited losses. Trading in options involves risk and is not suitable for all investors. Prior to buying or selling an option, investors must review the 'Characteristics and Risks of Standardized Options', at <https://www.theocc.com/company-information/documents-and-archives/options-disclosure-document>. If you are unable to access the website, please contact your Deutsche Bank representative for a copy of this important document.

Participants in foreign exchange transactions may incur risks arising from several factors, including the following: (i) exchange rates can be volatile and are subject to large fluctuations; (ii) the value of currencies may be affected by numerous market factors, including world and national economic, political and regulatory events, events in equity and debt markets and changes in interest rates; and (iii) currencies may be subject to devaluation or government-imposed exchange controls, which could affect the value of the currency. Investors in securities such as ADRs, whose values are affected by the currency of an underlying security, effectively assume currency risk.

Unless governing law provides otherwise, all transactions should be executed through the Deutsche Bank entity in the investor's home jurisdiction. Aside from within this report, important conflict disclosures can also be found at <https://research.db.com/Research/> on each company's research page or under the 'Disclosures' tab. Investors are strongly encouraged to review this information before investing.

Deutsche Bank (which includes Deutsche Bank AG, its branches and affiliated companies) is not acting as a financial adviser, consultant or fiduciary to you or any of your agents (collectively, "You" or "Your") with respect to any information provided in this report. Deutsche Bank does not provide investment, legal, tax or accounting advice, Deutsche Bank is not acting as your impartial adviser, and does not express any opinion or recommendation whatsoever as to any strategies, products or any other information presented in the materials. Information contained herein is being provided solely on the basis that the recipient will make an independent assessment of the merits of any investment decision, and it does not constitute a recommendation of, or express an opinion on, any product or service or any trading strategy.



The information presented is general in nature and is not directed to retirement accounts or any specific person or account type, and is therefore provided to You on the express basis that it is not advice, and You may not rely upon it in making Your decision. The information we provide is being directed only to persons we believe to be financially sophisticated, who are capable of evaluating investment risks independently, both in general and with regard to particular transactions and investment strategies, and who understand that Deutsche Bank has financial interests in the offering of its products and services. If this is not the case, or if You are an IRA or other retail investor receiving this directly from us, we ask that you inform us immediately.

In July 2018, Deutsche Bank revised its rating system for short term ideas whereby the branding has been changed to Catalyst Calls ("CC") from SOLAR ideas; the rating categories for Catalyst Calls originated in the Americas region have been made consistent with the categories used by Analysts globally; and the effective time period for CCs has been reduced from a maximum of 180 days to 90 days.

United States: Approved and/or distributed by Deutsche Bank Securities Incorporated, a member of FINRA and SIPC. Analysts located outside of the United States are employed by non-US affiliates and are not registered/qualified as research analysts with FINRA.

European Economic Area (exc. United Kingdom): Approved and/or distributed by Deutsche Bank AG, a joint stock corporation with limited liability incorporated in the Federal Republic of Germany with its principal office in Frankfurt am Main. Deutsche Bank AG is authorized under German Banking Law and is subject to supervision by the European Central Bank and by BaFin, Germany's Federal Financial Supervisory Authority.

United Kingdom: Approved and/or distributed by Deutsche Bank AG acting through its London Branch at 21 Moorfields, London EC2Y 9DB. Deutsche Bank AG in the United Kingdom is authorised by the Prudential Regulation Authority and is subject to limited regulation by the Prudential Regulation Authority and Financial Conduct Authority. Details about the extent of our authorisation and regulation are available on request.

Hong Kong SAR: Distributed by Deutsche Bank AG, Hong Kong Branch, except for any research content relating to futures contracts within the meaning of the Hong Kong Securities and Futures Ordinance Cap. 571. Research reports on such futures contracts are not intended for access by persons who are located, incorporated, constituted or resident in Hong Kong. The author(s) of a research report, and the entities for which they act and/or are affiliated with, may not be licensed to carry on regulated activities in Hong Kong, and if not licensed, do not hold themselves out as being able to do so. The provisions set out above in the 'Additional Information' section shall apply to the fullest extent permissible by local laws and regulations, including without limitation the Code of Conduct for Persons Licensed or Registered with the Securities and Futures Commission. This report is intended for distribution only to 'professional investors' as defined in Part 1 of Schedule of the SFO. This document must not be acted or relied on by persons who are not professional investors. Any investment or investment activity to which this document relates is only available to professional investors and will be engaged only with professional investors.

India: Prepared by Deutsche Equities India Private Limited (DEIPL) having CIN: U65990MH2002PTC137431 and registered office at 14th Floor, The Capital, C-70, G Block, Bandra Kurla Complex, Mumbai (India) 400051. Tel: + 91 22 7180 4444. It is registered by the Securities and Exchange Board of India (SEBI) as a Stock broker bearing registration no.: INZ000252437; Merchant Banker bearing SEBI Registration no.: INM000010833 and Research Analyst bearing SEBI Registration no.: INH000001741. DEIPL's Compliance / Grievance officer is Ms. Rashmi Poddar (Tel: +91 22 7180 4929 email ID: complaints.deipl@db.com). Registration granted by SEBI and certification from NISM in no way guarantee performance of DEIPL or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. DEIPL may have received administrative warnings from the SEBI for breaches of Indian regulations. Deutsche Bank and/or its affiliate(s) may have debt holdings or positions in the subject company. With regard to information on associates, please refer to the "Shareholdings" section in the Annual Report at: <https://www.db.com/ir/en/annual-reports.htm>. For latest India research audit report, refer https://country.db.com/india/deutsche-equities-india/index?language_id=1.

Japan: Approved and/or distributed by Deutsche Securities Inc.(DSI). Registration number - Registered as a financial instruments dealer by the Head of the Kanto Local Finance Bureau (Kinsho) No. 117. Member of associations: JSDA,



Type II Financial Instruments Firms Association and The Financial Futures Association of Japan. Commissions and risks involved in stock transactions - for stock transactions, we charge stock commissions and consumption tax by multiplying the transaction amount by the commission rate agreed with each customer. Stock transactions can lead to losses as a result of share price fluctuations and other factors. Transactions in foreign stocks can lead to additional losses stemming from foreign exchange fluctuations. We may also charge commissions and fees for certain categories of investment advice, products and services. Recommended investment strategies, products and services carry the risk of losses to principal and other losses as a result of changes in market and/or economic trends, and/or fluctuations in market value. Before deciding on the purchase of financial products and/or services, customers should carefully read the relevant disclosures, prospectuses and other documentation. 'Moody's', 'Standard Poor's', and 'Fitch' mentioned in this report are not registered credit rating agencies in Japan unless Japan or 'Nippon' is specifically designated in the name of the entity. Reports on Japanese listed companies not written by analysts of DSI are written by Deutsche Bank Group's analysts with the coverage companies specified by DSI. Some of the foreign securities stated on this report are not disclosed according to the Financial Instruments and Exchange Law of Japan. Target prices set by Deutsche Bank's equity analysts are based on a 12-month forecast period.

Korea: Distributed by Deutsche Securities Korea Co.

South Africa: Deutsche Bank AG Johannesburg is incorporated in the Federal Republic of Germany (Branch Register Number in South Africa: 1998/003298/10).

Singapore: This report is issued by Deutsche Bank AG, Singapore Branch (One Raffles Quay #18-00 South Tower Singapore 048583, 65 6423 8001), which may be contacted in respect of any matters arising from, or in connection with, this report. Where this report is issued or promulgated by Deutsche Bank in Singapore to a person who is not an accredited investor, expert investor or institutional investor (as defined in the applicable Singapore laws and regulations), they accept legal responsibility to such person for its contents.

Taiwan: Information on securities/investments that trade in Taiwan is for your reference only. Readers should independently evaluate investment risks and are solely responsible for their investment decisions. Deutsche Bank research may not be distributed to the Taiwan public media or quoted or used by the Taiwan public media without written consent. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation to trade in such securities/instruments.

Qatar: Deutsche Bank AG in the Qatar Financial Centre (registered no. 00032) is regulated by the Qatar Financial Centre Regulatory Authority. Deutsche Bank AG - QFC Branch may undertake only the financial services activities that fall within the scope of its existing QFCRA license. Its principal place of business in the QFC: Qatar Financial Centre, Tower, West Bay, Level 5, PO Box 14928, Doha, Qatar. This information has been distributed by Deutsche Bank AG. Related financial products or services are only available only to Business Customers, as defined by the Qatar Financial Centre Regulatory Authority.

Russia: The information, interpretation and opinions submitted herein are not in the context of, and do not constitute, any appraisal or evaluation activity requiring a license in the Russian Federation.

Kingdom of Saudi Arabia: Deutsche Securities Saudi Arabia (DSSA) is a closed joint stock company authorized by the Capital Market Authority of the Kingdom of Saudi Arabia with a license number (No. 37-07073) to conduct the following business activities: Dealing, Arranging, Advising, and Custody activities. DSSA registered office is Faisaliah Tower, 17th Floor, King Fahad Road - Al Olaya District Riyadh, Kingdom of Saudi Arabia P.O. Box 301806.

United Arab Emirates: Deutsche Bank AG in the Dubai International Financial Centre (registered no. 00045) is regulated by the Dubai Financial Services Authority. Deutsche Bank AG - DIFC Branch may only undertake the financial services activities that fall within the scope of its existing DFSA license. Principal place of business in the DIFC: Dubai International Financial Centre, The Gate Village, Building 5, PO Box 504902, Dubai, U.A.E. This information has been distributed by Deutsche Bank AG. Related financial products or services are available only to Professional Clients, as defined by the Dubai Financial Services Authority.



Australia and New Zealand: This research is intended only for 'wholesale clients' within the meaning of the Australian Corporations Act and New Zealand Financial Advisors Act, respectively. Please refer to Australia-specific research disclosures and related information at https://www.dbresearch.com/PROD/RPS_EN-PROD/PROD000000000521304.xhtml. Where research refers to any particular financial product recipients of the research should consider any product disclosure statement, prospectus or other applicable disclosure document before making any decision about whether to acquire the product. In preparing this report, the primary analyst or an individual who assisted in the preparation of this report has likely been in contact with the company that is the subject of this research for confirmation/clarification of data, facts, statements, permission to use company-sourced material in the report, and/or site-visit attendance. Without prior approval from Research Management, analysts may not accept from current or potential Banking clients the costs of travel, accommodations, or other expenses incurred by analysts attending site visits, conferences, social events, and the like. Similarly, without prior approval from Research Management and Anti-Bribery and Corruption ("ABC") team, analysts may not accept perks or other items of value for their personal use from issuers they cover.

Additional information relative to securities, other financial products or issuers discussed in this report is available upon request. This report may not be reproduced, distributed or published without Deutsche Bank's prior written consent.

Backtested, hypothetical or simulated performance results have inherent limitations. Unlike an actual performance record based on trading actual client portfolios, simulated results are achieved by means of the retroactive application of a backtested model itself designed with the benefit of hindsight. Taking into account historical events the backtesting of performance also differs from actual account performance because an actual investment strategy may be adjusted any time, for any reason, including a response to material, economic or market factors. The backtested performance includes hypothetical results that do not reflect the reinvestment of dividends and other earnings or the deduction of advisory fees, brokerage or other commissions, and any other expenses that a client would have paid or actually paid. No representation is made that any trading strategy or account will or is likely to achieve profits or losses similar to those shown. Alternative modeling techniques or assumptions might produce significantly different results and prove to be more appropriate. Past hypothetical backtest results are neither an indicator nor guarantee of future returns. Actual results will vary, perhaps materially, from the analysis.

The method for computing individual E,S,G and composite ESG scores set forth herein is a novel method developed by the Research department within Deutsche Bank AG, computed using a systematic approach without human intervention. Different data providers, market sectors and geographies approach ESG analysis and incorporate the findings in a variety of ways. As such, the ESG scores referred to herein may differ from equivalent ratings developed and implemented by other ESG data providers in the market and may also differ from equivalent ratings developed and implemented by other divisions within the Deutsche Bank Group. Such ESG scores also differ from other ratings and rankings that have historically been applied in research reports published by Deutsche Bank AG. Further, such ESG scores do not represent a formal or official view of Deutsche Bank AG.

It should be noted that the decision to incorporate ESG factors into any investment strategy may inhibit the ability to participate in certain investment opportunities that otherwise would be consistent with your investment objective and other principal investment strategies. The returns on a portfolio consisting primarily of sustainable investments may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered, and the investment opportunities available to such portfolios may differ. Companies may not necessarily meet high performance standards on all aspects of ESG or sustainable investing issues; there is also no guarantee that any company will meet expectations in connection with corporate responsibility, sustainability, and/or impact performance.

Copyright © 2026 Deutsche Bank AG



David Folkerts-Landau

Group Chief Economist and Global Head of Research

Pam Finelli
COO and Head of Fixed
Income Research

Steve Pollard
Global Head of Company
Research and Sales

Jim Reid
Global Head of Macro and
Thematic Research

Tim Rokossa
Head of European
Company Research

Matthew Barnard
Head of Americas
Company Research

Debbie Jones
Global Head of
Sustainability and Data
Innovation, Research

Robin Winkler
Head of German Macro
Research

Sameer Goel
Global Head of EM &
APAC Research

Francis Yared
Global Head of Rates
Research

George Saravelos
Global Head of FX
Research

Peter Hooper
Vice-Chair of Research

Nilendra de-Mel
Head of APAC & Middle
East Product
Development

International Production Locations

Deutsche Bank AG
Deutsche Bank Place
Level 16
Corner of Hunter & Phillip
Streets
Sydney, NSW 2000
Australia
Tel: (61) 2 8258 1234

Deutsche Bank AG
Equity Research
Mainzer Landstrasse 11-17
60329 Frankfurt am Main
Germany
Tel: (49) 69 910 00

Deutsche Bank AG
Filiale Hongkong
International Commerce
Centre
1 Austin Road West,
Kowloon,
Hong Kong
Tel: (852) 2203 8888

Deutsche Securities Inc.
1-3-1 Azabudai
Azabudai Hills Mori JP
Tower
Minato-ku, Tokyo 106-
0041
Japan
Tel: (81) 3 6730 1000

Deutsche Bank AG
21 Moorfields
London EC2Y 9DB
United Kingdom
Tel: (44) 20 7545 8000

**Deutsche Bank Securities
Inc.**
The Deutsche Bank Center
1 Columbus Circle
New York, NY 10019
Tel: (1) 212 250 2500

Deutsche Bank AG
Filiale Singapur
One Raffles Quay, South
Tower
Singapore 048583
Tel: (65) 6423 8001